



United International University

Department of Computer Science and Engineering

CSE 4495: Software Testing and Quality Assurance

Final Examination: Summer 2025

Total Marks: 40 Time: 2 hour

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all questions. The numbers on the right of the questions denote their marks.

1. (a) Many shops use, **UBuyU** a common platform to digitize their tasks of browsing, ordering items and the payment method. Recently, **UBuyU** has been facing two major problems:
- Some orders are placed by users who were not active at that time.
 - The whole payment method sometimes crashes when users send very long messages with special symbols or emojis.

Which two quality attributes are affected most by these problems? [CO1 – 4]

- (b) The university's online class platform is running with some issues. During live video classes, students complain that the video *keeps stopping and buffering*. But other features, like file downloading and assignment submission, are working fine.

According to the students, they cannot access the system every night from 02:00 am to 04:00 am which hampers their preparation.

The rule says the system must meet these targets:

Availability $\geq 96\%$, POFOD < 0.1 & RCOF < 0.3 per hour

- i. **Which performance and reliability metrics should be measured to understand issues shared in the first paragraph?** [CO1 – 2]

- ii. **Does the system meet or dissatisfy any of the targets?** [CO1 – 2]

- (c) Discuss the differences between static and dynamic verifications. [CO1 – 2]

2. A hospital's appointment system lets patients book slots with doctors. A booking is accepted if the doctor is on duty in that slot and no other patient has already booked the same doctor-date-hour slot.

The patient might be booking the doctor for the first time, or they might be for a follow-up appointment. Also, the patient can be with or without a reference which becomes invalid if the patient is for a follow-up appointment. On success, the system returns a confirmation code. Otherwise, it rejects with a specific reason.

- (a) Identify your choices while testing this feature. Choices are aspects that you have control over that can alter the output of the program. Also, enumerate representative values for your choices. [CO2 – 6]

- (b) Use Category Partition Method (CPM) to reduce the total number of test cases. [CO3 – 6]

3. The Product class models an item with name, details, price, and a maximum discount limit. The *setMaximumDiscountRate()* method updates the maximum discount but throws an **IllegalArgumentException** if the rate is **negative or above 100%**. The *getDiscountedPrice(requestedRate)* method returns the **price after applying the smaller of the requested and maximum discount rates**. The *getDetails()* method fetches product information from the internet but must complete within one second.

Now, write JUnit Test functions to check:

Product
- name: String - details: String - price: Double - maxDiscountRate: Double
+ Product(name, price, maxDiscountRate) + getMaximumDiscountRate() + setMaximumDiscountRate() + getDiscountedPrice(requestedRate) + getDetails()

Figure 1: Product Class

- (a) whether the *setMaximumDiscountRate* and *getDiscountedPrice* methods work correctly. [CO3 – 4]
(b) whether the *getDetails* method returns data (a String) within the given time limit. [CO3 – 4]

4. Given the Java code below :

```
public static int compute(int x, int limit, boolean reset) {  
    if (x < 0) return -1;  
  
    int sum = 0;  
    int i = 0;  
  
    do {  
        if (x % 2 == 0) {  
            sum += x;  
        } else if (x % 3 == 0) {  
            sum += 3;  
        } else {  
            sum += 1;  
        }  
  
        if (reset && sum > limit) {  
            sum = 0;  
        }  
  
        i++;  
    } while (i < 3 && sum < limit);  
  
    return sum;  
}
```

- (a) Draw the control-flow graph (CFG) for **compute** method. [CO4 – 3]
(b) Provide a *minimal* set of concrete tests that achieves **100% branch coverage**. [CO4 – 4]
(c) Briefly explain, with a small example, whether **Modified Condition/ Decision Coverage (MC/DC)** subsumes **branch coverage** and whether it subsumes **path coverage**. [CO4 – 3]